

Mathematical Methods for International Commerce

Week 12/2: Definite Integration (6.2)

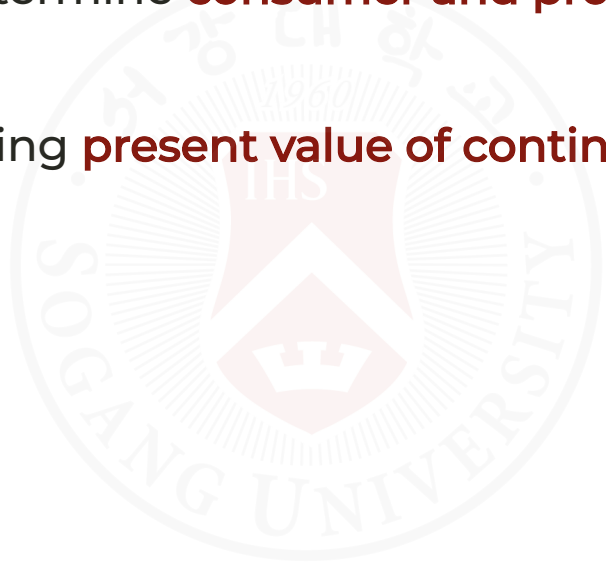
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Why It Matters in Economics & Finance

- Definite integration is crucial for calculating areas under curves, total costs, revenues, and investment values.
- In economics, it is used to determine **consumer and producer surplus** and **capital accumulation**.
- In finance, it helps in calculating **present value of continuous revenue streams**.



Understanding Definite Integration

- Integration finds the **area under a curve** between two specific points.
 - The definite integral gives the net signed area between the curve and the x-axis.
- Notation: If $f(x)$ is continuous over $[a, b]$, then:

$$\int_a^b f(x) dx = F(b) - F(a)$$

where

- $f(x)$ is the function to be integrated,
- a and b are the limits of integration,
- $F(x)$ is the **antiderivative** of $f(x)$.
 - $F(b)$ and $F(a)$ are the values of the antiderivative at b and a , respectively.

Visualizing the Definite Integral

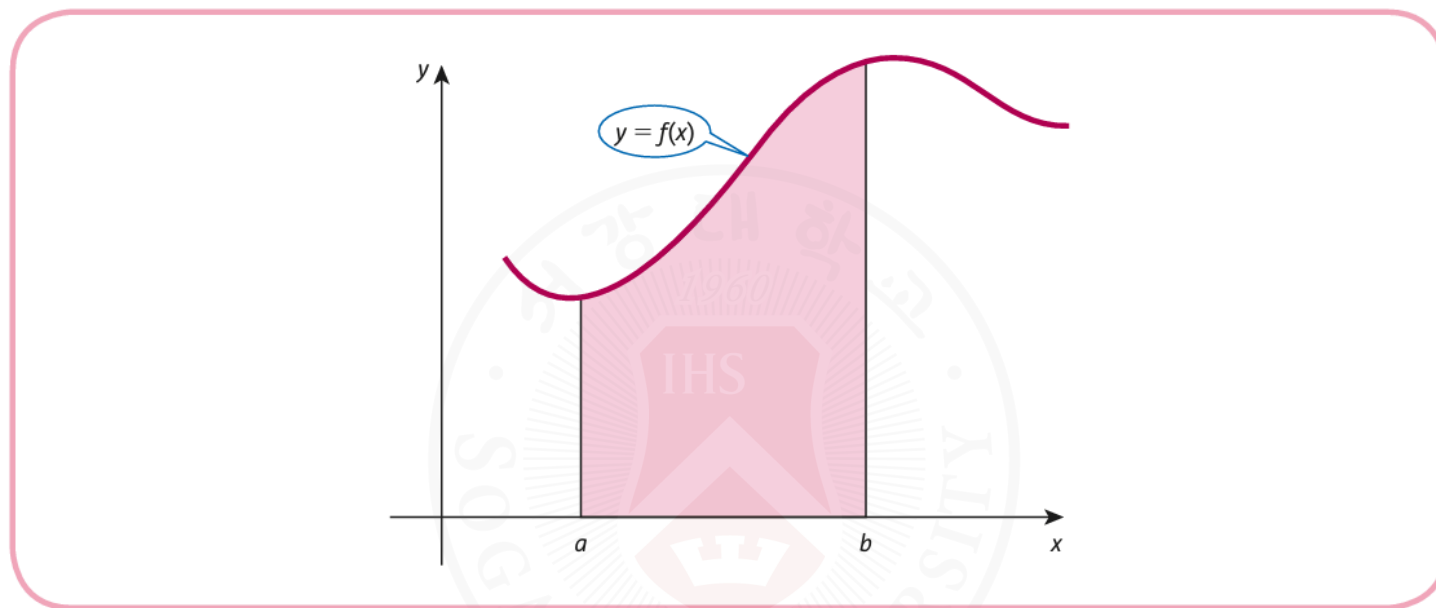


Figure 6.2

- This graph shows the general concept of finding the area under a curve between two points, a and b .
- The definite integral gives the **net area** between the curve and the x-axis over the interval $[a, b]$.
- The area can be positive or negative depending on the position of the curve relative to the x-axis.
- If the curve is above the x-axis, the area is positive; if below, the area is negative.
- If the curve crosses the x-axis, positive and negative areas partially offset each other.

Understanding Definite Integration (cont)

- Example:

$$\int_0^3 (2x + 1) dx = [x^2 + x]_0^3 = (9 + 3) - (0 + 0) = 12$$

- Example:

- Find the area under $y = x^2$ from $x = 1$ to $x = 2$.

$$\int_1^2 x^2 dx = \left[\frac{x^3}{3} \right]_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3}$$

Visualizing the Area Under a Curve

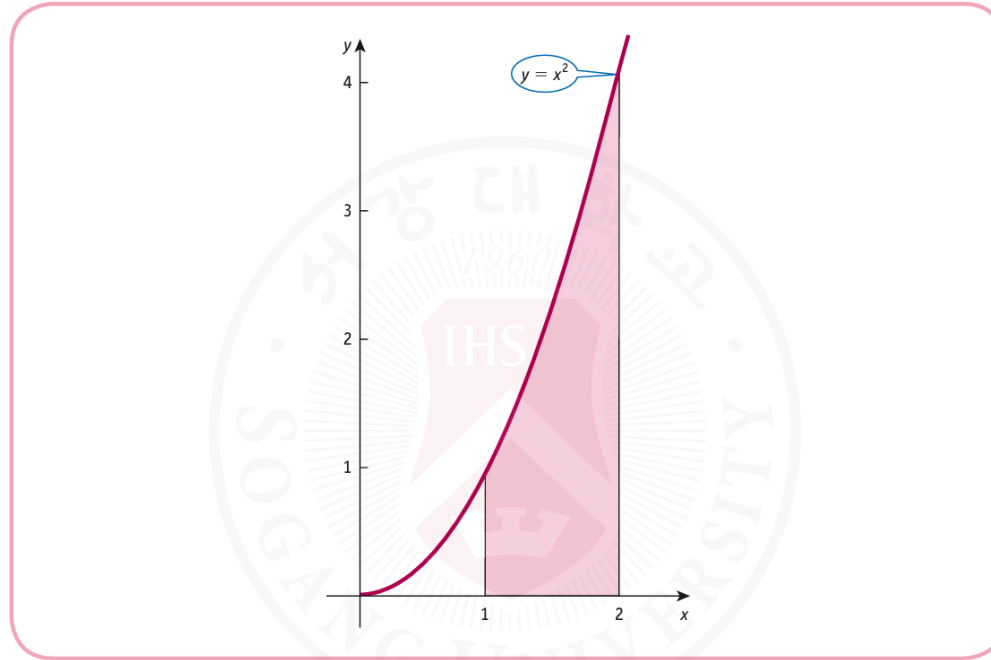


Figure 6.1

- This graph illustrates the area under the curve $y = x^2$ between $x = 1$ and $x = 2$.
- The area represents the **definite integral** value.

Definite Integration Examples

Example (a): Evaluating the Definite Integral

Evaluate the definite integral:

$$\int_2^6 3 \, dx$$

Solution:

We integrate the constant function:

$$\int_2^6 3 \, dx = [3x]_2^6 = 3(6) - 3(2) = 18 - 6 = 12$$

This can also be confirmed graphically by calculating the area of the rectangle (Figure 6.3):

- **Base:** $6 - 2 = 4$
- **Height:** 3
- **Area:** $4 \times 3 = 12$

Example (a): Evaluating the Definite Integral (cont)

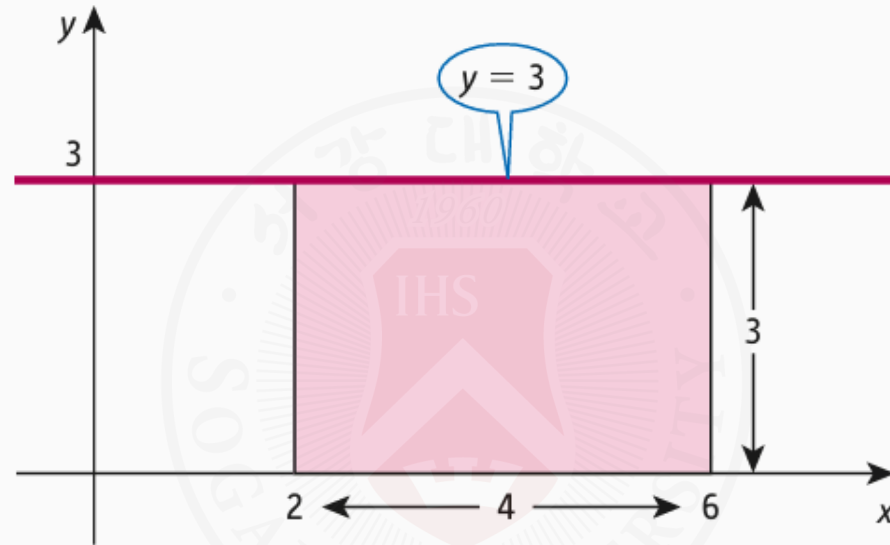


Figure 6.3

Example (b): Evaluating the Definite Integral

Evaluate the definite integral:

$$\int_0^2 (x + 1) dx$$

Solution:

We integrate the linear function:

$$\int_0^2 (x + 1) dx = \left[\frac{x^2}{2} + x \right]_0^2$$

Substituting the limits:

$$\begin{aligned} &= \left(\frac{2^2}{2} + 2 \right) - \left(\frac{0^2}{2} + 0 \right) \\ &= (2 + 2) - (0) = 4 \end{aligned}$$

Example (b): Evaluating the Definite Integral (cont)

Graphically, this is represented as a region under $y = x + 1$ (Figure 6.4a) and a one-half of the rectangle (Figure 6.4b):

- **Base:** 2
- **Height:** 4
- **Area:** $\frac{1}{2} \times 2 \times 4 = 4$

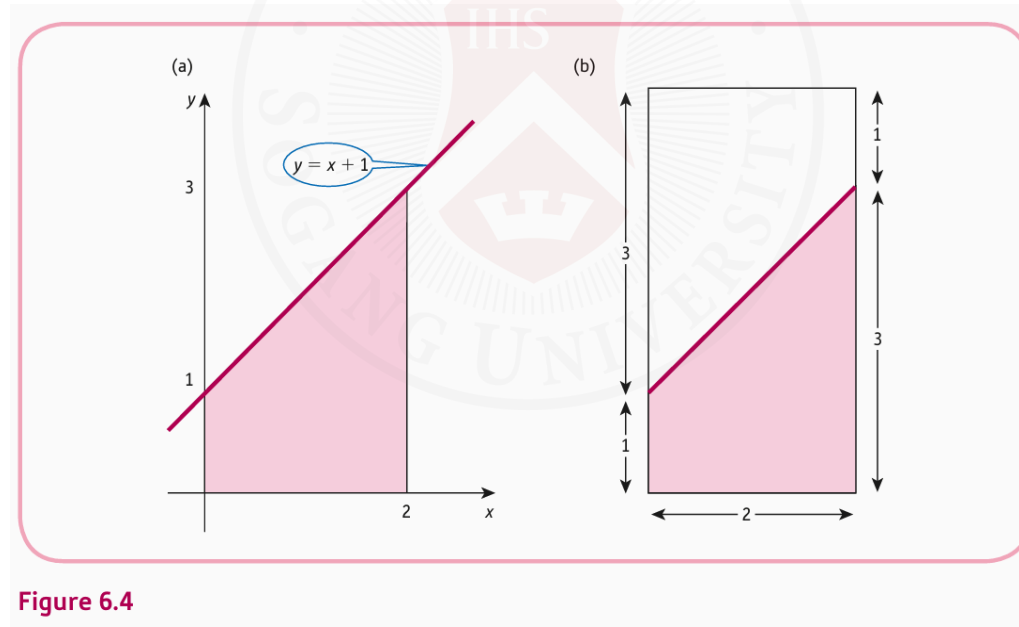


Figure 6.4

Area Under Demand Curve - Consumer Surplus

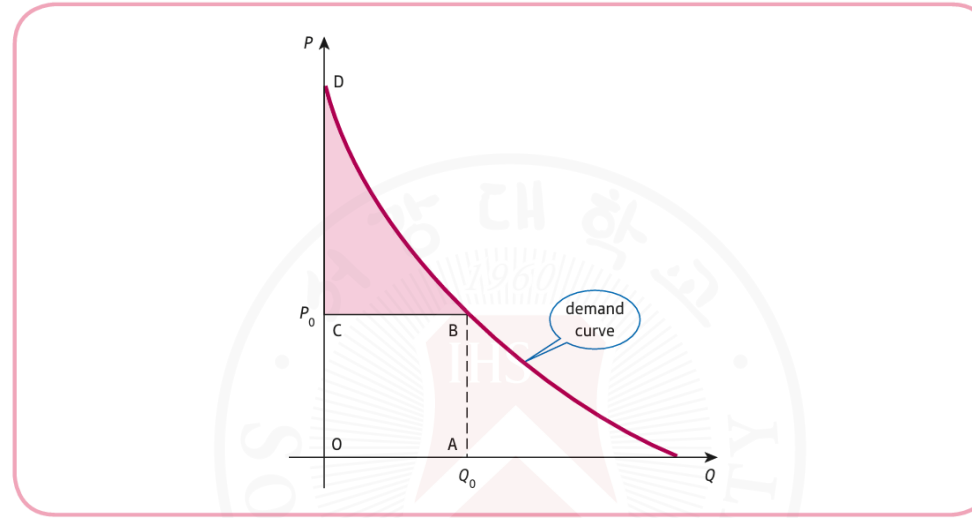


Figure 6.5

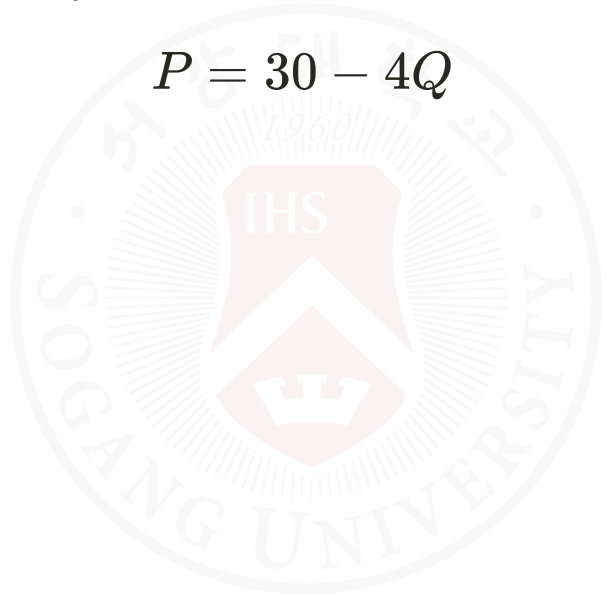
- Consumer surplus is the area under the demand curve and above the price line.
- It measures the **benefit consumers receive** when purchasing a product at a given price.

Area Under Demand Curve - Consumer Surplus (cont)

Problem Statement

- Find the **consumer's surplus** at $Q = 5$ for the demand function:

$$P = 30 - 4Q$$



Area Under Demand Curve - Consumer Surplus (cont)

Step 1: Define the Function

- The demand function is:

$$f(Q) = 30 - 4Q$$

- At $Q_0 = 5$, the price is:

$$P_0 = 30 - 4(5) = 10$$

Area Under Demand Curve - Consumer Surplus (cont)

Step 2: Consumer Surplus Formula

The formula for **consumer's surplus** is:

$$CS = \int_0^{Q_0} f(Q) dQ - Q_0 \times P_0$$

| Consumer surplus = area under the demand curve – actual amount paid.

Substituting the given values:

$$CS = \int_0^5 (30 - 4Q) dQ - 5 \times 10$$

Area Under Demand Curve - Consumer Surplus (cont)

Step 3: Solve the Integral with R

```
# Define the function
f <- function(Q) { 30 - 4 * Q }

# Integrate
cs_integral <- integrate(f, lower = 0, upper = 5)$value

# Calculate CS
P0 <- 10
Q0 <- 5
CS <- cs_integral - Q0 * P0

# Display result
CS
```

```
## [1] 50
```



Area Under Demand Curve - Consumer Surplus (cont)

Step 4: Mathematical Solution

1. Integrate:

$$\begin{aligned}\int_0^5 (30 - 4Q) dQ &= [30Q - 2Q^2]_0^5 \\ &= (30 \times 5 - 2 \times 5^2) - (30 \times 0 - 2 \times 0^2) \\ &= (150 - 50) - (0 - 0) = 100\end{aligned}$$

1. Calculate CS:

$$CS = 100 - 5 \times 10 = 100 - 50 = 50$$

Answer: The consumer surplus is **50 units**.

Producer Surplus

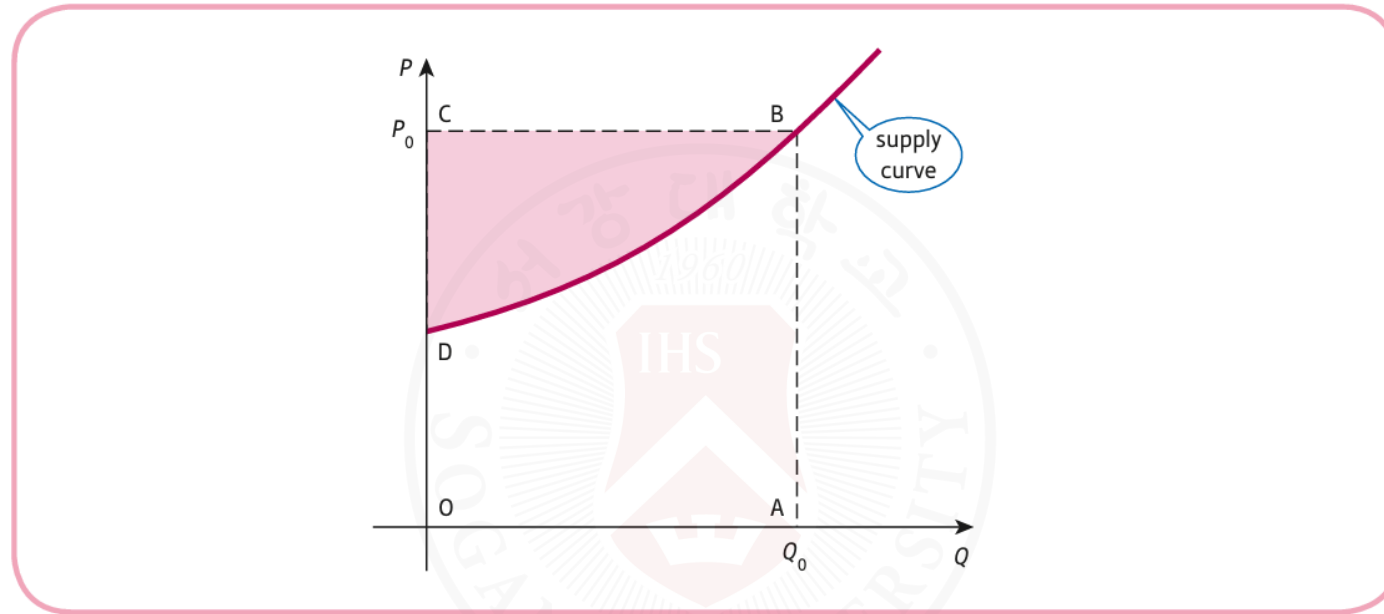


Figure 6.6

- Producer surplus is the area above the supply curve and below the price line.
- It represents the **benefit producers receive** by selling at a market price higher than the minimum they would accept.

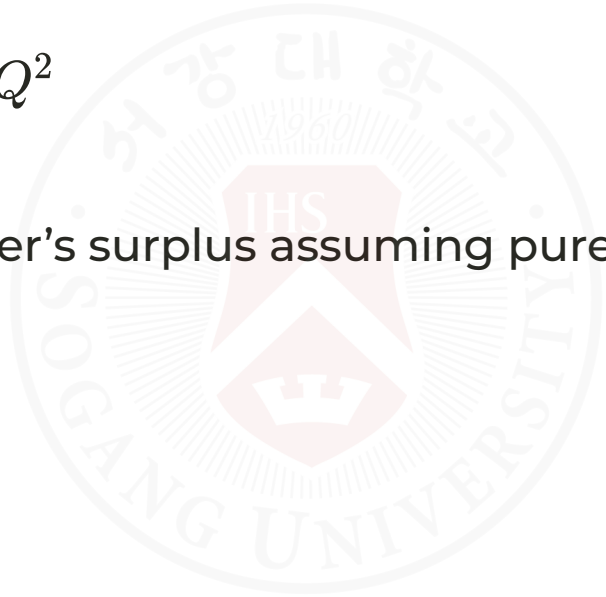
Producer Surplus (cont)

Producer's Surplus - Example

Given:

- Demand function: $P = 35 - Q^2$
- Supply function: $P = 3 + Q^2$

Objective: Calculate the producer's surplus assuming pure competition.



Producer Surplus (cont)

Step 1: Find Equilibrium

- Set demand equal to supply:

$$35 - Q^2 = 3 + Q^2$$

$$35 - 2Q^2 = 3$$

$$-2Q^2 = -32$$

$$Q^2 = 16 \implies Q = 4$$

- Equilibrium price:

$$P_0 = 35 - (4)^2 = 19$$

We take the positive root only because quantity cannot be negative.

Producer Surplus (cont)

Step 2: Define Producer Surplus

$$PS = Q_0 P_0 - \int_0^{Q_0} (3 + Q^2) dQ$$

Where:

- $Q_0 = 4$
- $P_0 = 19$



Producer Surplus (cont)

Step 3: Calculate Producer Surplus with R

```
Q0 <- 4
P0 <- 19

# Define the supply function
supply <- function(Q) {
  3 + Q^2
}

# Calculate integral
integral <- integrate(supply, lower = 0, upper = Q0)$value

# Calculate PS
PS <- Q0 * P0 - integral
PS
```

```
## [1] 42.66667
```

Producer Surplus (cont)

Step 4: Mathematical Solution

1. Calculate the integral:

$$\begin{aligned}\int_0^4 (3 + Q^2) dQ &= \left[3Q + \frac{Q^3}{3} \right]_0^4 \\ &= \left(3 \times 4 + \frac{4^3}{3} \right) - (0) \\ &= \left(12 + \frac{64}{3} \right) - 0 \\ &= \frac{36}{3} + \frac{64}{3} = \frac{100}{3}\end{aligned}$$

Producer Surplus (cont)

Step 4: Mathematical Solution (cont)

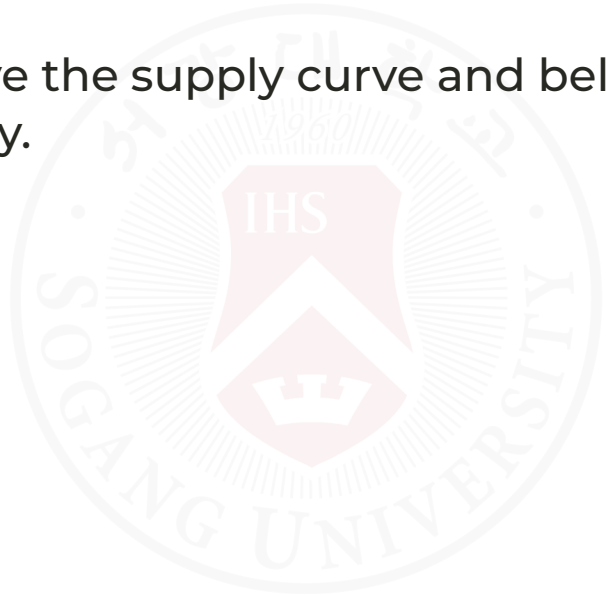
1. Calculate PS:

$$\begin{aligned} PS &= Q_0 P_0 - \int_0^{Q_0} (3 + Q^2) dQ \\ &= 4 \times 19 - \frac{100}{3} \\ &= 76 - \frac{100}{3} \\ &= \frac{228}{3} - \frac{100}{3} = \frac{128}{3} \\ &= 42.67 \end{aligned}$$

Producer Surplus (cont)

Step 5: Interpretation

- The producer's surplus is approximately 42.67.
- This represents the area above the supply curve and below the equilibrium price line up to the equilibrium quantity.



Net Investment and Capital Formation

Understanding Net Investment

- **Net Investment (I)** equals the rate of change of capital (K) stock over time
- The relationship is given by:

$$I(t) = \frac{dK}{dt}$$

- Here, $I(t)$ is the flow of money in dollars per year, and $K(t)$ is the accumulated capital.

Net Investment and Capital Formation (cont)

Capital Formation - Definite Integral

- To calculate the capital formation over a period from t_1 to t_2 , we use:

$$\int_{t_1}^{t_2} I(t) dt$$

- If we are given the investment flow function, we integrate to find the capital stock.

Note: Capital accumulation is the total amount of capital accumulated over time, considering continuous inflows and outflows.

Net Investment and Capital Formation (cont)

Example - Capital Formation

Investment Flow Function:

$$I(t) = 9000\sqrt{t}$$

(a) Calculate the capital formation from the end of the first year to the end of the fourth year with R:

```
# Integration for capital formation
capital_formation <- integrate(function(t) 9000 * sqrt(t), lower = 1, upper = 4)$value
capital_formation
```

```
## [1] 42000
```

Net Investment and Capital Formation (cont)

Example - Capital Formation (cont)

(b) Determine the number of years before capital stock exceeds \$100,000 with R.

We need to solve:

$$\int_0^T 9000\sqrt{t} dt = 100000$$

```
# Solving for T
solve_T <- function(T) {
  integral_value <- integrate(function(t) 9000 * sqrt(t), lower = 0, upper = T)$value
  return(integral_value - 100000)
}

T_value <- uniroot(solve_T, c(0, 10))$root
T_value
```

```
## [1] 6.524767
```

- The capital stock reaches \$100,000 approximately **6.5 years** into the investment period.

Net Investment and Capital Formation (cont)

Example - Capital Formation (cont)

Mathematical Solution (a)

- The integral is:

$$\int_1^4 9000\sqrt{t} dt = 9000 \left[\frac{2}{3} t^{3/2} \right]_1^4$$

- Evaluating the integral:

$$\begin{aligned} &= 9000 \left(\frac{2}{3} (4^{3/2}) - \frac{2}{3} (1^{3/2}) \right) \\ &= 9000 \left(\frac{2}{3} (8) - \frac{2}{3} (1) \right) \\ &= 9000 \left(\frac{16}{3} - \frac{2}{3} \right) = 9000 \left(\frac{14}{3} \right) = 42000 \end{aligned}$$

Answer: The capital formation from the end of the first year to the end of the fourth year is **\$42,000**.

Net Investment and Capital Formation (cont)

Example - Capital Formation (cont)

Mathematical Solution (b)

- To find T such that:

$$\int_0^T 9000\sqrt{t} dt = 100000$$

- The integral is:

$$\int_0^T 9000\sqrt{t} dt = 9000 \left[\frac{2}{3} t^{3/2} \right]_0^T$$

- Evaluating the integral:

$$\begin{aligned} &= 9000 \left(\frac{2}{3} T^{3/2} - 0 \right) \\ &= 6000 T^{3/2} \end{aligned}$$

Net Investment and Capital Formation (cont)

Example - Capital Formation (cont)

Mathematical Solution (b) (cont)

- Setting equal to 100,000:

$$6000T^{3/2} = 100000$$

$$T^{3/2} = \frac{100000}{6000}$$

$$T^{3/2} = \frac{50}{3}$$

$$T = \left(\frac{50}{3}\right)^{2/3} \approx 6.5$$

Answer: The capital stock exceeds 100,000 approximately **6.5 years** after the start of the investment.

Net Investment and Capital Formation (cont)

Interpretation and Insights

- Capital formation provides a measure of how investment over time accumulates into capital stock.
- Understanding the time it takes for investment to achieve certain capital levels helps in financial planning and forecasting.



Present Value of Continuous Revenue Stream

Problem Statement

- Calculate the present value of a continuous revenue stream of \$1000 per year for 5 years, discounted at 9% annually.
- The present value is given by:

$$P = \int_0^5 1000e^{-0.09t} dt$$

Present Value of Continuous Revenue Stream (cont)

Step 1: Setup the Integral

- We need to evaluate the definite integral:

$$P = \int_0^5 1000e^{-0.09t} dt$$

- This integral represents the **present value** of a revenue stream discounted continuously.

Present Value of Continuous Revenue Stream (cont)

Step 2: Solve the Integral

- The integral is of the form:

$$\int e^{at} dt = \frac{1}{a} e^{at} + C$$

(for $a \neq 0$)

- Applying the integral:

$$\int 1000e^{-0.09t} dt = \frac{1000}{-0.09} e^{-0.09t} \Big|_0^5$$

Present Value of Continuous Revenue Stream (cont)

Step 3: Evaluate the Integral

- Evaluating at the bounds:

$$\begin{aligned} P &= \frac{1000}{0.09} (e^{-0.09 \times 5} - e^0) \\ &= -11111.11 (e^{-0.45} - 1) \end{aligned}$$

Present Value of Continuous Revenue Stream (cont)

Step 4: Calculate the Present Value

- Approximating $e^{-0.45} \approx 0.6376$:

$$\begin{aligned} P &\approx -11111.11(0.6376 - 1) \\ &\approx -11111.11 \times -0.3624 \\ &\approx 4026.35 \end{aligned}$$

Answer: The present value of the continuous revenue stream is **\$4026.35**.

Practice Problems

1. Evaluate the definite integral:

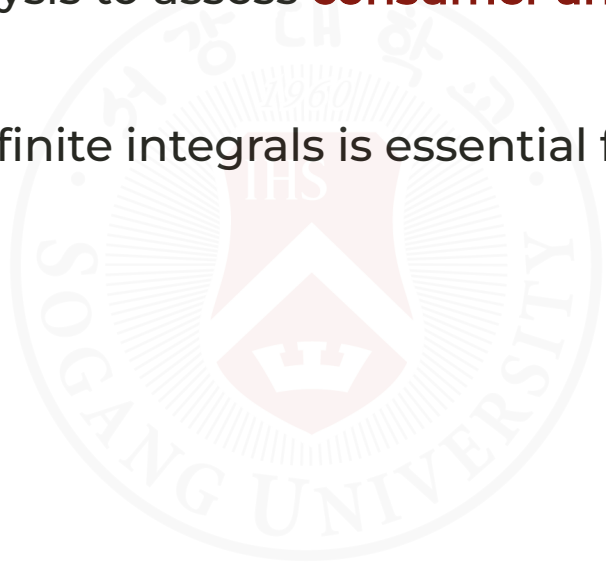
- $\int_1^4 (3x^2 + 2) dx$

2. Calculate the consumer surplus given the demand curve $P = 50 - 2Q$ and market price $P = 30$.

3. Determine the present value of a continuous revenue stream given by $R(t) = 150e^{0.04t}$ over 3 years, discounted at 6% annually.

Summary

- Definite integration calculates areas under curves, total costs, revenues, and surplus values.
- It is applied in economic analysis to assess **consumer and producer surplus** and **capital accumulation**.
- Setting up and evaluating definite integrals is essential for economic applications.



The seal of Gangneung University is a circular emblem. It features a central shield with a crown on top and the letters 'IHS' inside. The year '1960' is inscribed above the shield. The outer ring of the seal contains the university's name in Korean and English.

2. Home work #2

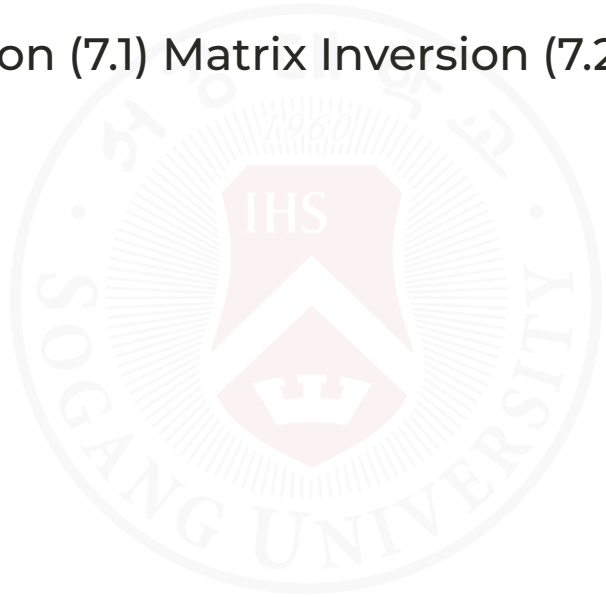
Homework #2

- **Due Date:** June 11, 2026, before the start of class.
- **Submission Format:** Submit your solutions as a single PDF file via the Cyber Campus.
- **Instructions:**
 - Clearly show all steps and calculations.
 - Include explanations for your answers where applicable.
 - Ensure your submission is neat and well-organized.
 - Bring any questions to the office hours or email me.
- Work on your Home Assignment #2 (Jacques, 10th edition, Chapters 5-9):
 - Chapter 5.1: Exercise 5.1, Problem 5 (p. 411)
 - Chapter 5.2: Exercise 5.2, Problem 7 (p. 427)
 - Chapter 5.3: Exercise 5.3, Practice Problem inside the chapter (p. 432 only)
 - Chapter 5.4: Exercise 5.4, Problem 5 (p. 456)
 - Chapter 5.5: Exercise 5.5, Problems 6 and 7 (p. 467)
 - Chapter 5.6: Exercise 5.6, Problem 4 (p. 479)
 - Chapter 6.1: Exercise 6.1, Problem 3 (p. 506)
 - Chapter 6.2: Exercise 6.2, Problem 6 (p. 521)
 - Chapter 7.1: Exercise 7.1, Problem 5 (p. 553)
 - Chapter 7.2: Exercise 7.2, Problem 6 (p. 572)
 - Chapter 7.3: Exercise 7.3, Problem 5 (p. 584)
 - Chapter 8.1: Exercise 8.1, Problem 6 (p. 615)
 - Chapter 8.2: Exercise 8.2, Problem 3 (p. 626)
 - Chapter 9.1: Exercise 9.1, Problem 4 (p. 655)
 - Chapter 9.2: Exercise 9.2, Problem 3 (p. 670)

Good luck!

Next Classes

- (May 26) Quiz #2
 - Covers materials after Midterm Exam
- (May 28) Basic Matrix Operation (7.1) Matrix Inversion (7.2)





Any QUESTIONS?

Thank you for your attention and participation!